A KRR Mechanism in In-Context Transformers

Fernando Ruiz Mazo

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1 The Question

A transformer can match kernel ridge regression (KRR) predictions on sampled labels without implementing the KRR rule. Pointwise agreement tests one vector y. A different computation can agree at that vector and disagree on nearby counterfactual labels.

We therefore freeze the input geometry $G = (X_{\text{ctx}}, X_{\text{tgt}})$ and study the label-response map. With G fixed, KRR is a linear operator T from context labels to target predictions. The first question is behavioral:

$$\text{does the transformer's local response } D_\varepsilon F(y)[u] \text{ match } Tu?$$

The answer is only interpreted on task-distributed label directions, not on arbitrary Euclidean perturbations.

The mechanistic question is sharper. If the transformer has a local KRR response, is that response supported by a compact context-side residual-stream subspace? For an activation-derived basis Q, the KRR-constrained reduced operator is $T_Q = K_t Q Q^\top$. We ask whether T_Q is prediction-equivalent to exact KRR at a predeclared posterior-risk tolerance, when such a subspace becomes available across layers, and whether removing KRR-important directions damages the model's response.

The claim is deliberately local. In the successful regimes, the transformer locally realizes the task-distributed KRR label-response operator, and this behavior is explained by low-dimensional activation subspaces whose required dimension is computed from the finite KRR task before looking at activations. The experiments do not claim global equality with KRR or an SGD theorem for learning the mechanism.

2 Fixed Geometry Makes Ridge Regression an Operator

Fix one in-context episode with context and target inputs

$$X_{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times d_x}, \qquad X_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d_x}.$$

The fixed geometry is

$$G = (X_{\text{ctx}}, X_{\text{tgt}}).$$

During the operator tests, G is held fixed. The context inputs do not move, the target inputs do not move, and the only varying object is the context-label vector

$$y = (y_1, \dots, y_{n_{\text{ctx}}}) \in \mathbb{R}^{n_{\text{ctx}}}.$$

Define

$$K = X_{\mathrm{ctx}} X_{\mathrm{ctx}}^{\top}, \qquad K_t = X_{\mathrm{tgt}} X_{\mathrm{ctx}}^{\top}, \qquad A = K + \sigma^2 I.$$

For linear-kernel ridge regression, the dual coefficients and target predictions are

$$\alpha^{\star} = A^{-1} y, \qquad y^{\star} = K_t \alpha^{\star} = K_t A^{-1} y.$$

Thus, once G is fixed, ridge regression is a linear map from context labels to target predictions:

$$T_G := K_t A^{-1}, \qquad T_G : \mathbb{R}^{n_{\mathrm{ctx}}} \to \mathbb{R}^{n_{\mathrm{tgt}}}.$$

The transformer defines a map on the same spaces:

$$F_\theta^G : \mathbb{R}^{n_{\mathrm{ctx}}} \to \mathbb{R}^{n_{\mathrm{tgt}}}, \qquad F_\theta^G(y) = \widehat{y}_\theta.$$

When the geometry is clear, write T and F. The central comparison is not merely $F(y) \approx Ty$ for one y. It is whether the local response of F to label perturbations matches the operator T.

3 Context-Label Directions, Local Operators, and Additivity

A context-label direction is a vector

$$u \in \mathbb{R}^{n_{\mathrm{ctx}}}.$$

It assigns one scalar to each context example. Perturbing in direction u means changing labels as

$$y + \varepsilon u = (y_1 + \varepsilon u_1, \ldots, y_{n_{\mathrm{ctx}}} + \varepsilon u_{n_{\mathrm{ctx}}}).$$

This is a direction over context positions. It is not a direction in input-feature space and not a direction in target space. If there are 47 context examples, these directions live in $\mathbb{R}^{47}$.

Pointwise agreement checks only the value of two maps at one label vector:

$$F(y) \approx Ty.$$

Since fixed-geometry ridge regression is linear, its Jacobian is the operator itself:

$$J_T(y) = T \quad \text{for every } y.$$

The transformer map F can be nonlinear in the labels. The weakest local version of "the same label-to-prediction rule" is therefore equality of first-order responses:

$$J_F(y) u \approx Tu.$$

Equivalently, for small label perturbations,

$$F(y + \delta) \approx F(y) + T\delta.$$

The experiments estimate the Jacobian action with the central finite difference

$$D_\varepsilon F(y)[u] := \frac{F(y + \varepsilon u) - F(y - \varepsilon u)}{2\varepsilon}.$$

The symmetric form is used because it cancels the leading first-order bias of a one-sided difference: for a smooth map, the error is $O(\varepsilon^2)$ rather than $O(\varepsilon)$. It also treats positive and negative label perturbations in the same direction symmetrically, which is important when the goal is to estimate the local tangent response rather than a one-sided effect. The behavioral operator test asks whether

$$D_\varepsilon F(y)[u] \approx Tu$$

on held-out label directions.

This is still a local statement. Matching the value and first derivative at one point does not imply global equality. For example, $f(x) = x + x^2$ and $g(x) = x$ have the same value and derivative at $x = 0$, but differ away from zero. Thus derivative matching supports tangent-level agreement, not $F(y') = Ty'$ for all possible labels.

The probe distribution specifies which counterfactual label directions are tested. The main probes are task-label probes:

$$u \sim \mathcal{N}(0, A).$$

This is the natural law because, under the matched linear task

$$y = X_{\text{ctx}}\beta + \eta, \qquad \beta \sim \mathcal{N}(0, I), \qquad \eta \sim \mathcal{N}(0, \sigma^2 I),$$

conditioning on the fixed geometry leaves only β and η random, so

$$y \mid G \sim \mathcal{N}(0, X_{\text{ctx}} X_{\text{ctx}}^\top + \sigma^2 I) = \mathcal{N}(0, A).$$

Equivalently, the conditional label covariance is

$$\operatorname{Cov}(y \mid G) = A.$$

Thus $u \sim \mathcal{N}(0, A)$ samples counterfactual label perturbations with the same covariance geometry as labels generated by the task, after the input geometry has been fixed. In this precise sense, task probes are the directions the task distribution itself emphasizes. In whitened coordinates, they are $u = A^{1/2} z$, $z \sim \mathcal{N}(0, I)$. The experiments also use isotropic probes

$$u \sim \mathcal{N}(0, I),$$

which test arbitrary context-label directions equally. The distinction is central: the claim is task-local because the transformer is close to ridge on task-label probes but not on arbitrary isotropic probes. Build probes are used to construct subspaces or fitted controls; held-out probes are used for reported errors.

For a held-out probe set $U = \{u_1, \ldots, u_m\}$ and a fixed linear operator $S : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}$, define

$$E_\varepsilon(F, S; U) := \left(\frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - S u_j\|_2^2}{\sum_{j=1}^m \|T u_j\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

When $S = T$, this measures whether the transformer's finite-difference response matches exact ridge regression. The denominator normalizes by the ridge-response energy on the same probes, so an error of 0.01 means a root-mean-square discrepancy of about one percent of the typical ridge response size.

Because ridge is linear, its higher-order derivatives are zero. We do not try to estimate all higher-order derivatives of the transformer. Instead, we check whether nonlinear interactions are small at the tested perturbation scale. Ridge has

$$T(u + v) = Tu + Tv.$$

A nonlinear transformer need not satisfy

$$D_\varepsilon F(y)[u+v] \approx D_\varepsilon F(y)[u] + D_\varepsilon F(y)[v].$$

If this approximate additivity failed badly, then $E_\varepsilon(F,T)$ would mix two effects: failure to be ridge and failure to have a stable local linear response.

For paired directions $V = \{(u_j, v_j)\}_{j=1}^m$, define the second mixed finite difference

$$C_j = F(y + \varepsilon u_j + \varepsilon v_j) - F(y + \varepsilon u_j - \varepsilon v_j) - F(y - \varepsilon u_j + \varepsilon v_j) + F(y - \varepsilon u_j - \varepsilon v_j),$$

and the additivity defect

$$A_\varepsilon(F;V) := \left(\frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j + \varepsilon v_j)\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

For any affine function of the labels, $C_j = 0$ exactly. Because the defect is normalized by ridge-response energy, values below one mean the non-additive interaction is smaller than the typical ridge response on the tested paired probes; values around one mean it is ridge-scale; values above one mean the interaction is larger than the ridge response, so a local-linear interpretation is unreliable. This diagnostic does not identify the operator. It only says whether a local operator comparison is meaningful at the chosen ε and probe law.

4 Activation Subspaces and Prediction-Risk Rank

The previous section gives behavioral certificates:

$$F(y) \approx Ty, \qquad D_\varepsilon F(y)[u] \approx Tu.$$

These are necessary, but they are still input-output facts. They do not say whether the transformer represents the directions needed for ridge regression, or whether those directions are used. The mechanistic question is: does the residual stream expose a compact set of context-position directions sufficient for the prediction-relevant ridge computation?

This is the right internal object to look for because ridge uses context-position dual variables. For fixed geometry,

$$\alpha^\star(y) = A^{-1}y \in \mathbb{R}^{n_{\text{ctx}}}, \qquad Ty = K_t \alpha^\star(y).$$

The dual coordinate α_i is indexed by the i-th context example. The context residual stream has the same context axis. At layer ℓ,

$$H_\ell^{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times D},$$

where D is the residual width. Rows are context tokens, and the k-th column $H_\ell^{\text{ctx}}[:,k]$ is a scalar pattern over context positions, hence a vector in $\mathbb{R}^{n_{\text{ctx}}}$. Linear combinations of residual-stream columns therefore define candidate directions in the same space as ridge dual variables.

Let

$$Q = [q_1, \ldots, q_r] \in \mathbb{R}^{n_{\text{ctx}} \times r}$$

be a basis matrix for such candidate context-position directions. The subspace is the column span

$$\text{span}(Q) = \{Qc : c \in \mathbb{R}^r\} \subseteq \mathbb{R}^{n_{\text{ctx}}},$$

so Q is both a matrix representation and, by abuse of language, the subspace it spans. We use an A-orthonormal basis,

$$Q^\top AQ = I,$$

because the quadratic part of the ridge dual objective is $\frac{1}{2}\alpha^\top A\alpha$. Thus the natural size of a dual displacement is $\alpha^\top A\alpha$, and two directions are orthogonal in the ridge geometry when their cross-term in this quadratic energy vanishes.

Now define what it means for $\mathrm{span}(Q)$ to be useful for ridge. This is a restriction of the dual solution, not an ordinary restriction of the input label vector. Full ridge solves

$$\alpha^\star(y) = \arg\min_{\alpha\in\mathbb{R}^{n_{\mathrm{ctx}}}} \left\{ \frac{1}{2}\alpha^\top A\alpha - \alpha^\top y \right\},$$

whose first-order condition is $A\alpha - y = 0$, hence $\alpha^\star = A^{-1}y$. To restrict the dual solution to $\mathrm{span}(Q)$, write $\alpha = Qc$. Plugging this into the same dual objective gives

$$\frac{1}{2}c^\top Q^\top AQc - c^\top Q^\top y.$$

Since $Q^\top AQ = I$, this is

$$\frac{1}{2}c^\top c - c^\top Q^\top y.$$

Differentiating with respect to c gives $c - Q^\top y = 0$, so

$$c^\star = Q^\top y.$$

Therefore the restricted dual solution is

$$\alpha_Q(y) = QQ^\top y.$$

Without the A-orthonormal convention, the same calculation would give

$$\alpha_Q(y) = Q(Q^\top AQ)^{-1}Q^\top y.$$

Predictions are still read out through the context-target kernel matrix K_t, so the reduced ridge operator induced by Q is

$$T_Q := K_t QQ^\top.$$

This is the bridge from residual-stream directions to an operator-level certificate. We are not fitting an arbitrary probe. We are asking whether the directions exposed by the residual stream are enough so that ridge, restricted to those directions, still predicts like full ridge.

The reason this sufficiency test is prediction-relevance weighted rather than global is that not every label direction matters equally for prediction. Conditional on the fixed geometry,

$$y \mid G \sim \mathcal{N}(0, A).$$

So the task naturally emphasizes perturbations

$$u \sim \mathcal{N}(0, A),$$

not arbitrary Euclidean directions. A direction is task-relevant when it is likely under this covariance geometry and when ridge produces a non-negligible target response along it. This leads to a purely ridge-theoretic benchmark: before looking inside the transformer, ask how many directions are needed by the best possible reduced ridge response on task-distributed labels.

To turn this into a prediction-scale benchmark, we need a reference risk. The target quantity is not the noisy context-label vector y. It is the vector of latent, noise-free function values at the target inputs. Write

$$f_{\mathrm{tgt}} = \big(f(x_1^{\mathrm{tgt}}), \ldots, f(x_{n_{\mathrm{tgt}}}^{\mathrm{tgt}})\big) \in \mathbb{R}^{n_{\mathrm{tgt}}},$$

and let K_{tt} be the target-target kernel matrix. In the linear case, $K_{tt} = X_{\mathrm{tgt}} X_{\mathrm{tgt}}^\top$. Under the matched GP/KRR model,

$$\begin{bmatrix} y \\ f_{\mathrm{tgt}} \end{bmatrix} \,\Big|\, G \sim \mathcal{N}\left(0, \begin{bmatrix} A & K_t^\top \\ K_t & K_{tt} \end{bmatrix}\right),$$

because $y = f_{\mathrm{ctx}} + \eta$ includes label noise while f_{tgt} is the noise-free target function. Conditioning this Gaussian gives

$$\mathbb{E}[f_{\mathrm{tgt}} \mid y, G] = K_t A^{-1} y = Ty$$

and

$$\mathrm{Cov}(f_{\mathrm{tgt}} \mid y, G) = K_{tt} - K_t A^{-1} K_t^\top.$$

Thus exact KRR is the posterior mean predictor of the latent target values. It still has nonzero risk, because the noisy context labels do not determine the target function values exactly. Define this irreducible KRR posterior prediction risk by

$$R_{\mathrm{KRR}}(G) := \mathbb{E}\|f_{\mathrm{tgt}} - Ty\|_2^2 = \mathrm{tr}\Big(K_{tt} - K_t A^{-1} K_t^\top\Big),$$

where the expectation is conditional on the fixed geometry G.

Now compare exact KRR with any linear approximation S to T. Decompose

$$f_{\mathrm{tgt}} - Sy = \big(f_{\mathrm{tgt}} - Ty\big) + \big(T - S\big)y.$$

The first term is the posterior residual after exact KRR, and the second term is a function of the observed labels y. Since $Ty = \mathbb{E}[f_{\mathrm{tgt}} \mid y, G]$, the posterior residual is orthogonal to every function of y. Therefore the cross-term vanishes and

$$\mathbb{E}\|f_{\mathrm{tgt}} - Sy\|_2^2 = R_{\mathrm{KRR}}(G) + \mathbb{E}_{y\sim\mathcal{N}(0,A)}\|(T-S)y\|_2^2.$$

The second term is the extra prediction risk caused by using S instead of exact KRR. Thus the natural scale for deciding whether a reduced operator is good enough is not zero operator error, but excess error relative to the irreducible risk already incurred by exact KRR. Define

$$\rho_G(S) := \frac{\mathbb{E}_{y\sim\mathcal{N}(0,A)}\|(T-S)y\|_2^2}{R_{\mathrm{KRR}}(G) + \eta_{\mathrm{num}}}.$$

This asks how much prediction risk is added by using S instead of exact KRR.

The rank benchmark should also be defined before looking at activations. The right baseline is not zero error: exact KRR is already uncertain about the latent target values, with posterior risk $R_{\mathrm{KRR}}(G)$. Therefore a reduced operator should be called sufficient when its discarded KRR response adds only a small fraction of that unavoidable risk.

To make this intrinsic to the task distribution, put task-label perturbations in whitened coordinates:

$$u = A^{1/2} z, \qquad z \sim \mathcal{N}(0, I).$$

In these coordinates exact KRR acts as

$$Tu = K_t A^{-1} A^{1/2} z = K_t A^{-1/2} z.$$

Define the whitened KRR response operator

$$C_G := K_t A^{-1/2}.$$

This is not a different predictor; it is T written in coordinates where the label directions generated by the task are spherical. This is why the spectrum of C_G, not the raw Euclidean spectrum of T, is the relevant one for task-local difficulty.

Let

$$C_G = U \operatorname{diag}(s_i) V^\top, \qquad s_1 \geq s_2 \geq \cdots \geq 0.$$

The right singular vectors v_i are the task-label modes seen by KRR, and s_i^2 is the target response energy carried by mode i. Since z is isotropic,

$$\mathbb{E}\|C_G z\|_2^2 = \sum_i s_i^2.$$

Among all rank-r linear responses in these whitened coordinates, Eckart–Young says that the best one keeps the first r singular modes. Its unavoidable excess prediction error is exactly the discarded tail

$$\sum_{i>r} s_i^2.$$

Thus the best rank-r reduced KRR predictor has total prediction risk

$$R_{\mathrm{KRR}}(G) + \sum_{i>r} s_i^2.$$

The tail is compared to $R_{\mathrm{KRR}}(G)$ because the practical question is prediction equivalence: does removing the remaining modes add little error relative to the uncertainty that KRR cannot remove anyway?

Definition 1 (Prediction-risk rank)**.** *For fixed geometry G and tolerance $\alpha > 0$, define*

$$r_\alpha(G) := \min\left\{ r : \sum_{i>r} s_i^2 \leq \alpha R_{\mathrm{KRR}}(G) \right\}.$$

In the experiments below we write $r_T := r_{5\%}$, i.e. $\alpha = 0.05$. Thus r_T is the smallest rank for which the best reduced KRR response would have prediction risk at most

$$(1.05)\, R_{\mathrm{KRR}}(G).$$

It is not the algebraic rank of T, nor the number of nonzero singular values. It is the number of task-visible KRR modes needed to be prediction-equivalent to exact KRR at the chosen tolerance. Modes beyond r_T may be nonzero, but their combined contribution is below the risk budget.

This is a finite-episode benchmark. It is computed from the sampled matrices K, K_t, K_{tt} before looking at the transformer. The same definition applies to nonlinear kernels by replacing these with the corresponding finite kernel matrices; no population spectral assumption is required.

Activation subspaces are then tested against this benchmark. Let Q be an A-orthonormal candidate basis, $Q^\top AQ = I$, and set

$$C_G = K_t A^{-1/2}, \qquad W = A^{1/2}Q, \qquad P_Q = WW^\top.$$

Here W is the same context-side subspace expressed in whitened task-label coordinates. For $u = A^{1/2}z$,

$$Tu = C_G z, \qquad T_Q u = K_t QQ^\top A^{1/2} z = C_G P_Q z.$$

Therefore the prediction-risk excess of the reduced operator is

$$\rho_G(T_Q) = \frac{\|C_G(I - P_Q)\|_F^2}{R_{\mathrm{KRR}}(G) + \eta_{\mathrm{num}}}.$$

If W spans the top r right singular directions of C_G, then

$$\rho_G(T_Q) = \frac{\sum_{i>r} s_i^2}{R_{\mathrm{KRR}}(G) + \eta_{\mathrm{num}}}.$$

This formula is the bridge from r_T to activations. If $\dim \operatorname{span}(Q) < r_T$, the subspace cannot reach the five-percent prediction-risk tolerance. If $\dim \operatorname{span}(Q) \geq r_T$, success still requires alignment: $A^{1/2} \operatorname{span}(Q)$ must contain the leading task-label modes of C_G.

Remark 1 (Reading the numbers)**.** *The number $\rho_G(S)$ is an excess-risk ratio. Thus $\rho_G(S) = 0.05$ means that using S instead of exact KRR adds five percent of the KRR posterior prediction risk. Values near zero mean the reduced operator is prediction-equivalent to KRR at the chosen tolerance. Values near or above one mean that the reduction adds error comparable to, or larger than, the irreducible KRR risk. Transformer response errors $E_\varepsilon(F, T)$ and $E_\varepsilon(F, T_Q)$ are different finite-difference diagnostics and are not bounded by this risk scale.*

The mechanism argument therefore needs three distinct checks:

$$E_\varepsilon(F, T; U) \text{ small}, \qquad \rho_G(T_Q) \text{ small}, \qquad E_\varepsilon(F, T_Q; U) \text{ small}.$$

The first is behavioral. The second says the residual-stream subspace supports ridge in the prediction-risk sense. The third says the transformer's local response follows the reduced operator induced by that same subspace. The decomposition

$$D_\varepsilon F(y)[u] - Tu = \big(D_\varepsilon F(y)[u] - T_Q u\big) + \big(T_Q u - Tu\big)$$

keeps these claims separate.

It remains to describe how the residual-stream subspace is extracted. The experiments use two final-state constructions in this section, and the next section introduces the causal layerwise construction. Given any context-pattern matrix $M \in \mathbb{R}^{n_{\mathrm{ctx}} \times p}$, compute the A-weighted SVD

$$A^{1/2}M = U\Sigma V^\top,$$

keep the selected left singular directions, and map back:

$$Q(M) := A^{-1/2} U_{\mathrm{kept}}.$$

Then $Q(M)^\top AQ(M) = I$. The ordered prefixes of this basis define a rank curve. The prediction-risk rank tells how many ridge-relevant directions should be enough in principle; the experiments then ask where the residual-stream prefixes first achieve small $\rho_G(T_Q)$.

The raw final-state extractor uses the unperturbed final residual stream,

$$M_{\mathrm{raw}} = H_L^{\mathrm{ctx}}, \qquad Q_{\mathrm{raw}} := Q(M_{\mathrm{raw}}).$$

It asks an availability question: which context-position directions are linearly visible in the final residual stream, and are enough of them present to support the prediction-relevant ridge operator?

The response-only extractor uses finite-difference changes in the final residual stream,

$$Z_L(u) = \frac{H_L^{\mathrm{ctx}}(y + \varepsilon u) - H_L^{\mathrm{ctx}}(y - \varepsilon u)}{2\varepsilon}, \qquad M_{\mathrm{resp}} = [Z_L(u_1), \dots, Z_L(u_m)],$$

and sets

$$Q_{\mathrm{resp}} := Q(M_{\mathrm{resp}}).$$

It asks the stricter question: among the directions that actually move when labels are perturbed, is there a context-position subspace whose induced T_Q has small prediction-risk excess?

In both constructions Q is frozen after extraction. If the basis were re-extracted for each held-out perturbation, it would not define one stable reduced operator. The certificate is that a fixed subspace extracted from residual-stream activations induces a fixed T_Q that is prediction-risk sufficient and whose action matches the transformer's held-out local response.

5 Layerwise Availability Construction

The final-state certificate can show that a useful subspace is present at the end of the computation, but it does not explain how that subspace appears across layers. The revised layerwise construction is a post-training diagnostic for this question. It asks, at each layer, whether the layer's label-response span contains a subspace of approximately r_T dimensions whose induced reduced KRR operator reaches the prediction-risk tolerance.

This replaces the older primary emphasis on one-at-a-time causal direction selection. Individual ablation is a useful causal-use diagnostic, but it is not the cleanest definition of emergence: a distributed representation can contain the right subspace while no single arbitrary basis direction is individually necessary. The layerwise sufficiency construction first finds a KRR-sufficient subspace; causal intervention should then test the selected subspace as a whole.

At layer ℓ, define the hidden-state finite-difference response

$$Z_\ell(u) := \frac{H_\ell^{\mathrm{ctx}}(y + \varepsilon u) - H_\ell^{\mathrm{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stacking these matrices over build probes gives

$$M_\ell = [Z_\ell(u_1), \dots, Z_\ell(u_m)].$$

The columns of M_ℓ are context-position patterns that are visible at layer ℓ when context labels move. An A-weighted SVD proposes an A-orthonormal candidate response span

$$A^{1/2} M_\ell = U_\ell \Sigma_\ell V_\ell^\top, \qquad q_i = A^{-1/2} U_{\ell,i}.$$

Keeping the singular directions above a numerical threshold gives a candidate span C_ℓ. This span is activation-derived. The next step is deliberately KRR-targeted: given the directions that the transformer makes available at layer ℓ, we ask whether that span contains a KRR-sufficient subspace of the intrinsic task rank. This question is necessarily oracle-guided with respect to exact KRR, because “KRR-sufficient” is defined by the excess-risk metric ρ_G. We therefore order directions inside C_ℓ by KRR usefulness. If C_ℓ is represented by an A-orthonormal matrix, set

$$W_\ell = A^{1/2} C_\ell, \qquad Z = K_t A^{-1/2}.$$

The eigenvectors of

$$W_\ell^\top Z^\top Z W_\ell$$

ordered from largest to smallest define nested prefixes

$$Q_{\ell,0} \subset Q_{\ell,1} \subset \cdots \subset C_\ell.$$

This ordering does not claim that the transformer itself has chosen this basis. It is an availability test: among the response directions exposed by the transformer, how good is the best KRR-aligned prefix? For each prefix we form

$$T_{Q_{\ell,k}} = K_t Q_{\ell,k} Q_{\ell,k}^\top$$

and evaluate

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}})A^{1/2}\|_F^2}{R_{\mathrm{KRR}} + \eta_{\mathrm{num}}}.$$

The two ranks in the experiment have different meanings. The prediction-risk rank $r_T(G; 0.05)$ is intrinsic to the finite KRR task: it is computed from $C_G = K_t A^{-1/2}$ before looking at the transformer. It is the smallest optimal rank that leaves at most five percent of KRR posterior risk in the discarded tail. By contrast, the layerwise rank-to-five-percent

$$k_\ell^{5\%} := \min\{k : \rho_G(T_{Q_{\ell,k}}) \le 0.05\}$$

is representation-dependent: it asks how many directions from the transformer's candidate response span C_ℓ are needed to reach the same risk tolerance. If $k_\ell^{5\%} \approx r_T$, then the layer contains an almost optimally aligned KRR subspace. If $k_\ell^{5\%} \gg r_T$, the span contains relevant information but in a poorly aligned form. If no such k exists, that layer's candidate span is not sufficient. We also report $\rho_G(T_{Q_{\ell,r_T}})$, the excess risk of using exactly the intrinsic task rank, and the best excess achievable inside C_ℓ.

6 Experiment 1: Final-State Operator Certificate

The experiments are organized as an evidence ladder. Experiment 1 first certifies that the trained model has a task-local ridge-like response at the final output and that this response is explained by a compact final-state context subspace. Experiment 2 asks where such a subspace becomes available across layers. Experiment 3 then intervenes on the certified directions to test causal use. All checkpoint names below refer to the archived final checkpoints under `checkpoints/final/`; when useful, we also give the original training artifact.

We use the same metric conventions throughout the tables. $E(F, S)$ always denotes the finite-difference operator error between the transformer's local label response and a comparison operator S, normalized by exact-KRR response energy on the same held-out probes; subscripts such as E_{task} and E_{iso} specify the probe distribution. $\rho_G(S)$ is the exact KRR prediction-risk excess of S, normalized by the posterior KRR risk. Pointwise prediction errors are different: they use the unperturbed label vector y, for example $\|F(y) - Ty\|/\|Ty\|$. Thus E measures a local operator, while pointwise errors measure one episode's actual prediction.

The first step is deliberately narrow. For each episode, the geometry G is fixed, exact KRR is the operator $T = K_t A^{-1}$, and the transformer is probed only by infinitesimal context-label perturbations. Build probes are used to extract the raw and response final-state bases; all reported errors use held-out task-label probes after the basis is frozen.

The trained model is `linear_baseline_dx5_L8.pt`, copied from the original `checkpoints/model_L8.pt`. It is an 8-layer linear-regression ICL transformer with $d_x = 5$, residual width $D = 128$, 4 attention heads, and no layer normalization. The evaluation uses the linear kernel

$$K = X_{\mathrm{ctx}} X_{\mathrm{ctx}}^\top, \qquad K_t = X_{\mathrm{tgt}} X_{\mathrm{ctx}}^\top,$$

with $n_{\mathrm{ctx}} = 47$, $n_{\mathrm{tgt}} = 3$, and $\sigma^2 = 0.1$. Episode geometries are drawn from the same flat-spectrum linear family used by the baseline sampler: for each episode,

$$c \sim \mathrm{Unif}[1, 10], \qquad x_i \sim \mathcal{N}(0, cI_5), \qquad \beta \sim \mathcal{N}(0, I_5),$$

with $y_i = x_i^\top \beta + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, on context positions and noiseless teacher values on target positions. The reported run uses 4 episodes, 16 build probes, 32 held-out probes, and finite-difference scale $\varepsilon = 10^{-3}$.

Raw and response bases are built as ordered A-weighted SVD prefixes. Rather than imposing a fixed rank cap, we evaluate the rank-prefix curve and select the first prefix whose reduced operator reaches the prediction-risk tolerance,

$$\rho_G(T_Q) \leq 0.05.$$

This gives rank 6 for both raw and response bases in the reported run. The larger curve is still shown, so the reader can see where the subspace becomes sufficient and whether additional directions materially change the conclusion.

The endpoint certificate has three parts. First, the model's local response is already close to KRR on task-label directions. Second, the activation-derived subspace induces a KRR-structured operator $T_Q = K_t Q Q^\top$ whose prediction-risk excess is far below the five-percent tolerance. Third, this is not merely a decodability result: a same-span least-squares readout can fit the model response slightly better, but it is much less KRR-like. In Table 1, $k_{\rho \leq 5\%}$ is the first prefix rank whose $\rho_G(T_Q) \leq 0.05$. The actLR control uses the same activation coordinates $Q^\top u$ but learns a free least-squares readout to match the transformer's finite-difference responses; it therefore tests decodability without imposing the KRR readout $K_t Q$. Matched random subspaces are random A-orthonormal subspaces with the same rank as the extracted basis.

Figure 1 shows the rank diagnostic behind the compactness claim. The x-axis increases the number of A-weighted SVD prefix directions used to form Q. The dashed line marks the first rank at which the task-label reduced operator satisfies $\rho_G(T_Q) \leq 0.05$, averaged over raw and response bases. The task-label curves collapse by ranks 5–6: at that point the reduced operator is prediction-risk sufficient, and the model-to-T_Q finite-difference error is already at its plateau.

Figure 2 visualizes the same-span readout control. The free activation-low-rank readout can reduce model-response error, especially on task probes, but it pays for that flexibility in prediction-risk excess. The KRR-constrained T_Q is therefore the stronger certificate: the activation coordinates do not merely decode the model's response; they support the KRR readout.

The task-local nature of the certificate is also visible continuously. For this visualization we make the comparison deliberately probe-law matched. For each mixing value, we draw perturbations

$$u_\lambda = \sqrt{1-\lambda}\, u_{\mathrm{task}} + \sqrt{\lambda}\, u_{\mathrm{iso}}, \qquad \lambda \in [0, 1],$$

where $u_{\mathrm{task}} \sim \mathcal{N}(0, A)$ and $u_{\mathrm{iso}} \sim \mathcal{N}(0, I)$. Thus $\lambda = 0$ is the task-label probe law used for the certificate, while $\lambda = 1$ is the isotropic off-distribution probe law. We then build a fresh response

Table 1: Experiment 1 endpoint certificate, mean over 4 episodes. $E(F,S)$ is the held-out finite-difference operator error on task-label probes. $\rho_G(S)$ is exact prediction-risk excess relative to KRR.

Check	Result	Reading
Model vs. exact KRR	$E_{\text{task}}(F,T) = 0.01259$; pointwise discrepancy 0.01045	The checkpoint is locally KRR-like on task-label perturbations.
Raw final basis	$k_{\rho\le 5\%} = 6$, $\rho_G(T_Q) = 0.00439$, $E_{\text{task}}(F,T_Q) = 0.01218$	The final residual stream contains a highly sufficient KRR source subspace.
Response final basis	$k_{\rho\le 5\%} = 6$, $\rho_G(T_Q) = 0.01651$, $E_{\text{task}}(F,T_Q) = 0.01343$	The sufficient subspace is also visible in label-induced hidden-state motion.
Same-span readout control	raw: $E_{\text{task}}(F,T_{\text{actLR}}) = 0.00765$, $\rho_G(T_{\text{actLR}}) = 0.14676$	A free readout can fit the model, but the KRR-constrained readout is the one that remains KRR-like.
Matched random subspaces	$\rho_G(T_Q) \approx 50.3$ at raw rank, $\approx$ 23.9 at response rank	Sufficiency is not explained by rank alone.
Isotropic label probes	$E_{\text{iso}}(F,T) \approx 0.294$	The certificate is task-local, not a global equality with ridge regression.

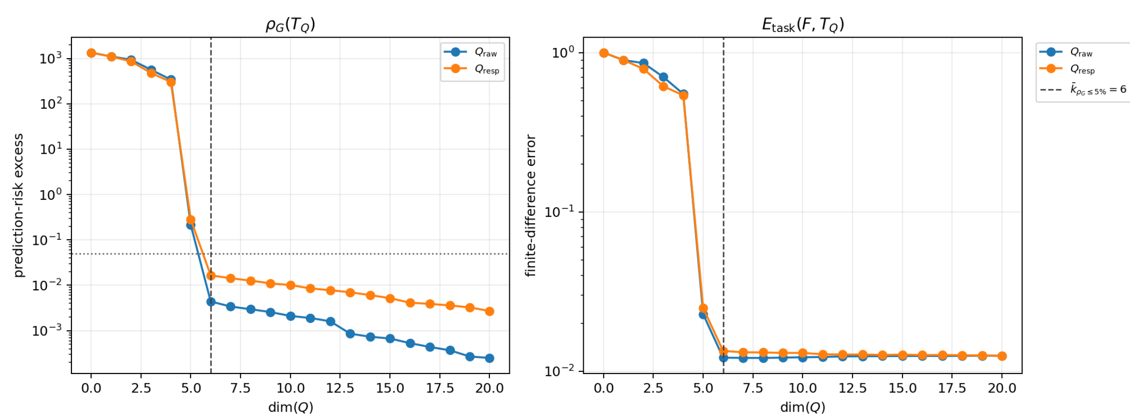

Figure 1: Experiment 1 rank-prefix curves. Left: prediction-risk excess $\rho_G(T_Q)$, with the dotted line marking the five-percent tolerance. Right: held-out task-probe response error $E_{\text{task}}(F,T_Q)$. The dashed vertical line marks the first prediction-risk-sufficient prefix, $k_{\rho\le 5\%} = 6$; the plotted ranks are prefixes of the same A-weighted SVD basis.

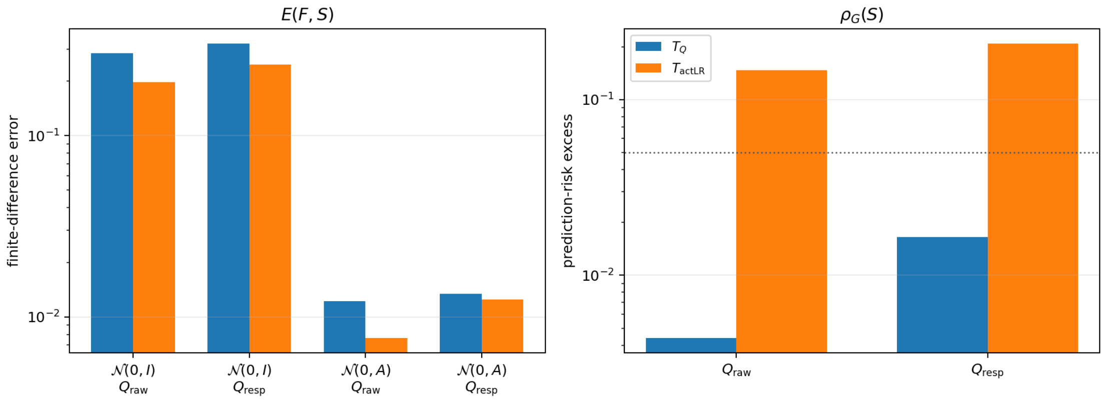

Figure 2: Experiment 1 same-span readout control. Left: finite-difference response error $E(F, S)$ on task and isotropic probes. Right: prediction-risk excess $\rho_G(S)$, shown once per basis because it is defined by the task covariance A rather than by the held-out probe law; the dotted line marks the five-percent tolerance. T_Q and actLR use the same activation coordinates $Q^\top u$. actLR fits the transformer's finite-difference responses with a free least-squares readout; T_Q uses the KRR-constrained readout $K_t Q$.

basis Q_λ from the same mixed hidden finite differences that are used for the plotted evaluation. This is not a held-out certificate; it is a fair matched-probe stress test asking whether the model's local KRR behavior persists as perturbations leave the task-distributed directions. Figure 3 shows the degradation directly in finite-difference response error.

The finite-difference comparison is meaningful because the local response is close to additive: at $\varepsilon = 10^{-3}$, the normalized additivity defect is 3.69×10^{-4} on task probes and 3.78×10^{-3} on isotropic probes. Thus Experiment 1 establishes the basic endpoint claim: on the task distribution, a compact activation subspace supports the KRR-constrained operator that the transformer locally follows.

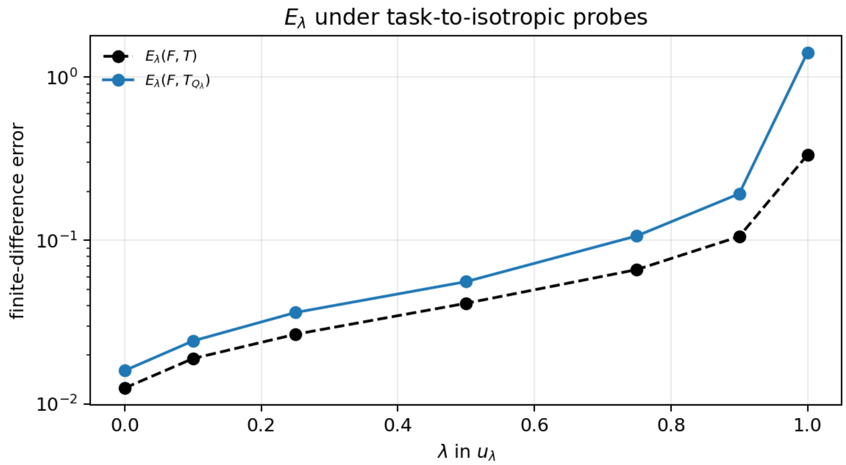

Figure 3: Experiment 1 task-to-isotropic probe interpolation with matched response bases. For each λ, Q_λ is built from the same mixed perturbations used in the plotted errors. The transformer is locally KRR-like on task-distributed directions, but the finite-difference operator error grows as the perturbations become isotropic.

7 Experiment 2: Prediction-Risk Rank and Layerwise Availability

Experiment 2 separates two facts that are easy to conflate. The prediction-risk rank r_T is an intrinsic property of the finite KRR problem; it is computed from $C_G = K_t A^{-1/2}$ before looking at the transformer. The layerwise rank $k_\ell^{5\%}$ is representation-dependent; it asks how many directions from the transformer's layer-ℓ response span are needed to reach $\rho_G \le 0.05$. The diagnostic therefore asks not whether the hidden state has dimension r_T, but whether it contains an approximately r_T-dimensional KRR-sufficient subspace.

At each layer, we form the response span C_ℓ from $M_\ell = D_y H_\ell^{\mathrm{ctx}}$. The span comes from activations, but the ordering inside it is KRR-targeted: we use the exact KRR operator to find the best KRR-aligned prefixes available inside that span. Thus Experiment 2 measures availability of a KRR-sufficient subspace, not whether the transformer autonomously selects this basis. Prefixes $Q_{\ell,k}$ are then evaluated by

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}})A^{1/2}\|_F^2}{R_{\mathrm{KRR}} + \eta_{\mathrm{num}}}.$$

In Experiment 2 tables, "cand. dim." is $\dim C_\ell$ after numerical truncation, "rank to 5%" is $k_\ell^{5\%}$, "best excess" is $\min_k \rho_G(T_{Q_{\ell,k}})$ over the tested prefixes, and "reach" is the fraction of evaluated episodes for which some prefix reaches $\rho_G \le 0.05$. A missing rank means the layer never reached the five-percent target in the tested prefix range.

The small linear checkpoints calibrate the diagnostic in the regime where the answer is expected to be simple. These trained models are `linear_sweep_dx3.pt`, `linear_sweep_dx5.pt`, `linear_sweep_dx8.pt`, `linear_sweep_dx10.pt`, and `linear_sweep_dx15.pt`; all use the same 8-layer, $D = 128$, 4-head architecture as Experiment 1, with the indicated input dimension d_x. For $d_x = 3, 5, 8, 10, 15$, with $n_{\mathrm{ctx}} = 47, n_{\mathrm{tgt}} = 16$, the prediction-risk rank matches the input dimension up to the target cap, and the final rank-to-five-percent matches r_T. This is not the main emergence result: in these linear tasks, the sufficient source subspace is already visible very early. Its role is to show that the revised r_T, $k_\ell^{5\%}$, and ρ_G diagnostic recovers the obvious low-dimensional answer.

Table 2: Experiment 2 linear calibration. The table checks that the prediction-risk diagnostic recovers $r_T \approx d_x$ in simple linear tasks. "First layer" is the first layer whose response span reaches $\rho_G \le 0.05$.

Checkpoint	r_T	first layer	final cand. dim.	$k_L^{5\%}$	$E_{\text{task}}(F,T)$
$d_x = 3$	3.0	0	27.5	3.0	0.00617
$d_x = 5$	5.0	0	27.6	5.0	0.00918
$d_x = 8$	8.0	0	40.1	8.0	0.01250
$d_x = 10$	10.0	0	40.4	10.0	0.01433
$d_x = 15$	14.9	2	46.6	14.9	0.01735

The cleanest emergence result is instead the nonlinear checkpoint `rbf_fixed_l3_nctx128_ntgt128_seed42.pt`, trained from scratch on the fixed RBF kernel with lengthscale 3.0, $d_x = 5$, $D = 128$, 8 layers, 4 heads, $n_{\text{ctx}} = 128$, and $n_{\text{tgt}} = 128$. We diagnose it in the same 128×128 regime. The model itself is locally close to exact KRR on task probes:

$$E_{\text{task}}(F,T) = 0.0306.$$

For this finite KRR problem, $r_T \approx 24.75$. Unlike the small linear sweep, the relevant subspace is not already present at layer 0. It appears over the first few layers.

Table 3: Experiment 2 main emergence result: selected layers of the RBF 128×128 checkpoint. Here $r_T \approx 24.75$. The response span is ordered by KRR usefulness before evaluating ρ_G.

Layer	cand. dim.	rank to 5%	$\rho_G(T_{Q_{\ell,r_T}})$
0	16	–	1.147
1	51	38.0	0.100
2	84	26.9	0.058
4	107	25.2	0.050
8	119	24.9	0.048

The transition is the main point. Layer 0 has only a 16-dimensional response span and is far from sufficient. By layer 2, a nearly task-rank subspace is visible; by layers 4 and 8, $k_\ell^{5\%}$ matches r_T to within finite-sample noise. The final response span is much larger than r_T, but it contains an approximately 25-dimensional KRR-sufficient subspace. Experiment 2 is therefore an availability diagnostic: it locates when the prediction-risk-rank subspace becomes linearly available, without yet claiming that the model causally uses it.

7.1 Stress Cases: Response Availability Tracks Failure

The $d_x = 40$ checkpoint `dx40_tau16_s122.pt` gives the cleanest warning against over-reading availability. It is a scale-aware 8-layer, $D = 128$, 4-head linear-regression checkpoint trained in a target-poor regime:

$$n_{\text{tgt}} = 8, \qquad \tau(\Sigma) = \sum_{i=1}^{40} \frac{\lambda_i}{\lambda_i + \sigma^2} \le 16,$$

with n_{ctx} sampled as a multiple of the effective dimension. The diagnostic instead uses

$$n_{\text{ctx}} = 47, \qquad n_{\text{tgt}} = 40,$$

so it asks for a richer simultaneous target operator than the checkpoint saw directly in one training episode. The checkpoint README reports

$$\text{MSE}/\text{MSE}_{\text{KRR}} = 1.638,$$

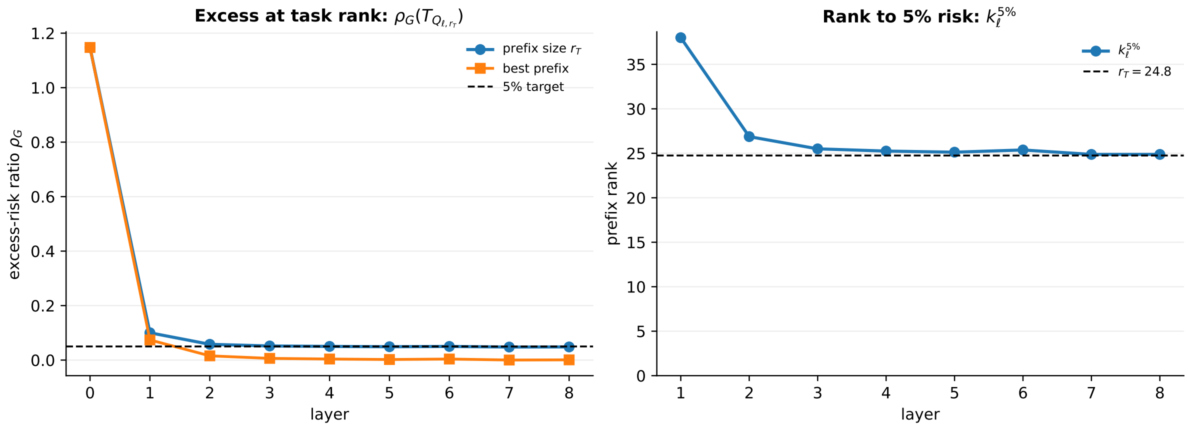

Figure 4: Experiment 2 layerwise emergence in the matched RBF 128×128 checkpoint. The candidate span is the response span $C_\ell = \mathrm{span}(D_y H_\ell^{\mathrm{ctx}})$. Left: the excess-risk ratio ρ_G for the prefix of size r_T and for the best prefix inside C_ℓ; the dashed line is the five-percent target. Right: the first prefix rank $k_\ell^{5\%}$ that reaches the target, with the intrinsic prediction-risk rank r_T shown as a dashed line. The disappearance of the large rank gap by the middle layers is the emergence signal.

so this checkpoint is not an exact-KRR solver. Since the layerwise diagnostic is meant to track the operator the transformer locally implements, the primary candidate span here is the response span $C_\ell = \mathrm{span}(D_y H_\ell^{\mathrm{ctx}})$: these are precisely the residual-stream directions activated to first order by context-label perturbations.

Table 4: $d_x = 40$ response-span stress test. Here $r_T = 21.0$ and $E_{\mathrm{task}}(F, T) = 0.0939$. "Reach" is the fraction of episodes whose response span contains some prefix with $\rho_G \le 0.05$.

Layer	cand. dim.	reach	$\rho_G(T_{Q_{\ell,r_T}})$	best ρ_G
0	16.0	0.00	0.965	0.965
2	35.0	0.50	0.164	0.137
5	38.0	0.50	0.105	0.073
8	40.0	0.50	0.118	0.086

The response span improves substantially over the first layers but does not become a reliable five-percent KRR response. At the final layer, using the task-rank prefix gives $\rho_G(T_{Q_{L,r_T}}) \approx 0.118$; even the best response-span prefix remains at $\rho_G \approx 0.086$, and only half the episodes reach the five-percent threshold at all. This is the mechanistic counterpart of the checkpoint's non-optimal prediction ratio: the model has learned a useful but imperfect target-rich KRR response.

Auxiliary raw-residual and union-span checks can find more KRR-aligned directions in this checkpoint, but we do not treat those as evidence for the implemented operator. They are containment checks: they show that some relevant information exists somewhere in the residual stream. The response span is the stricter diagnostic for what the model actually uses in the local label-to-output map.

We also reran the revised layerwise diagnostic on the older fixed-$\ell = 3$ RBF checkpoint `rbf_fixed_l3_original.pt`, copied from `checkpoints/model_rbf_fixed_l3.pt`, at $n_{\mathrm{ctx}} = 47$ and $n_{\mathrm{tgt}} = 16, 64, 128$. The final response-span summary is:

Thus this older checkpoint shows a softer version of the same lesson: response availability can improve with depth while the realized finite-difference operator remains less clean than in the

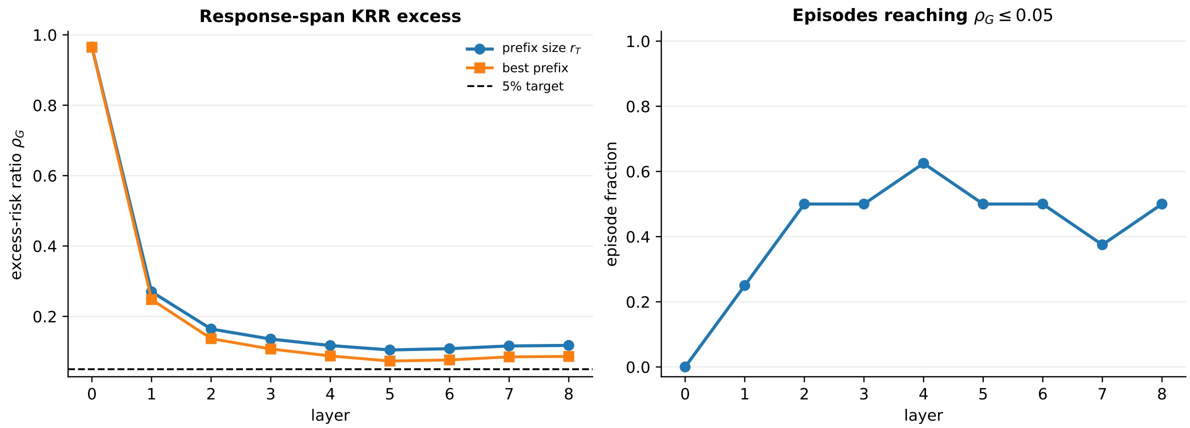

Figure 5: Experiment 2 $d_x = 40$ response-span stress case. Left: response-span excess risk at the prediction-risk rank r_T, together with the best prefix inside the response span. Both remain above the five-percent target. Right: fraction of episodes in which any response-span prefix reaches $\rho_G \leq 0.05$. The response geometry improves but never becomes a reliable prediction-risk-sufficient representation in the target-rich diagnostic regime.

Table 5: Older fixed-$\ell = 3$ RBF checkpoint, final-layer response diagnostic. The response span continues to contain a near-sufficient KRR subspace as n_{tgt} grows, but the realized finite-difference operator becomes less clean.

n_{tgt}	r_T	$k_L^{5\%}$	$\rho_G(T_{Q_{L,r_T}})$	$E_{\text{task}}(F,T)$
16	10.6	11.1	0.049	0.067
64	17.1	17.8	0.053	0.074
128	18.3	19.3	0.057	0.097

matched 128×128 RBF regime.

8 Experiment 3: Causal Direction Removal

The last step tests causal use. Experiments 1 and 2 show that a KRR-sufficient subspace is present; that still does not prove that the transformer relies on it. Experiment 3 therefore takes a certified raw final basis $Q = [q_1, \ldots, q_r]$, ranks its directions by KRR task leverage, and removes selected directions from the context-token residual stream before rolling the remaining layers forward.

Experiment 3 uses the same trained model as Experiment 1: `linear_baseline_dx5_L8.pt` (originally `model_L8.pt`). The diagnostic is run on flat-spectrum linear episodes with $d_x = 5$, $n_{\text{ctx}} = 47$, $n_{\text{tgt}} = 16$, $\sigma^2 = 0.1$, finite-difference scale $\varepsilon = 10^{-3}$, 32 episodes, and 32 held-out causal probes per episode. The complement-ablation run uses the same checkpoint and geometry.

The leverage score ranks directions by their effect on the KRR-constrained reduced operator, not by activation magnitude. For Q_{-j}, the basis with q_j removed, define

$$\Lambda_j^{\text{red}}(U) := \left(\frac{\sum_{u \in U} \|(T_Q - T_{Q_{-j}})u\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

High-leverage directions are the ones whose deletion most changes the reduced KRR response on task-label probes. The intervention removes selected directions at residual state s:

$$H_s^{\text{ctx}} \longmapsto H_s^{\text{ctx}} - \sum_{j \in J} q_j q_j^\top A H_s^{\text{ctx}},$$

and compares against matched controls: low-task-leverage directions from the same basis, high-isotropic-but-low-task directions, high-activation-variance-but-low-task directions, and random subsets. Residual state $s = 0$ is the embedding output, $s = i$ is the state after layer i, and $s = 8$ is the final residual state before the output head.

The Experiment 3 intervention metrics are normalized to be comparable across episodes. Prediction drift is $\|F_{\text{surg}}(y) - F_{\text{base}}(y)\| / \|F_{\text{base}}(y)\|$. KRR-error inflation is the post-surgery pointwise KRR error divided by the pre-surgery pointwise KRR error. Response damage is

$$\left(\frac{\sum_{u \in U} \|D_\varepsilon F_{\text{surg}}(y)[u] - D_\varepsilon F_{\text{base}}(y)[u]\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2},$$

so it measures how much the intervention changes the local finite-difference response, on the same task-label scale used for $E_{\text{task}}(F, T)$.

Before surgery, the checkpoint is in the task-local certificate regime:

$$E_{\text{task}}(F, T) = 0.00926, \qquad E_{\text{task}}(F, T_Q) = 0.00920, \qquad \rho_G(T_Q) = 0.00149.$$

The same episodes retain the off-distribution drift seen in Experiment 1:

$$E_{\text{iso}}(F, T) = 0.266.$$

The layer sweep adds localization: high-task surgery is most damaging at early residual states, before later attention has routed the relevant context information into target tokens, while context-only surgery after the final layer is a no-op. At $s = 1$, aggregate task leverage and task response damage are strongly related across matched non-random controls (log–log Pearson 0.91, Spearman 0.87).

Table 6: Experiment 3 causal-use summary. Direction removals are at residual state $s = 1$.

Intervention	Result	Reading
No surgery	$E_{\text{task}}(F, T) = 0.00926$, $\rho_G(T_Q) = 0.00149$	The starting episodes satisfy the same task-local certificate.
Remove one high-task-leverage direction	prediction drift 4.68, response damage 41.59, KRR-error inflation 649×	Directions selected by KRR leverage are causally important.
Remove one low-task-leverage direction	prediction drift 9.8×10^{-4}, response damage 0.0024, inflation 1.03×	Removing same-basis directions with low task leverage is nearly inert.
Keep only the Q-projection	$E_{\text{task}}(F, T) = 0.01836$, response damage 0.01634	The certified subspace nearly preserves the task-local response by itself.
Remove Q	$E_{\text{task}}(F, T) = 106.46$, response damage 106.46	The complement does not carry the task-local KRR response.

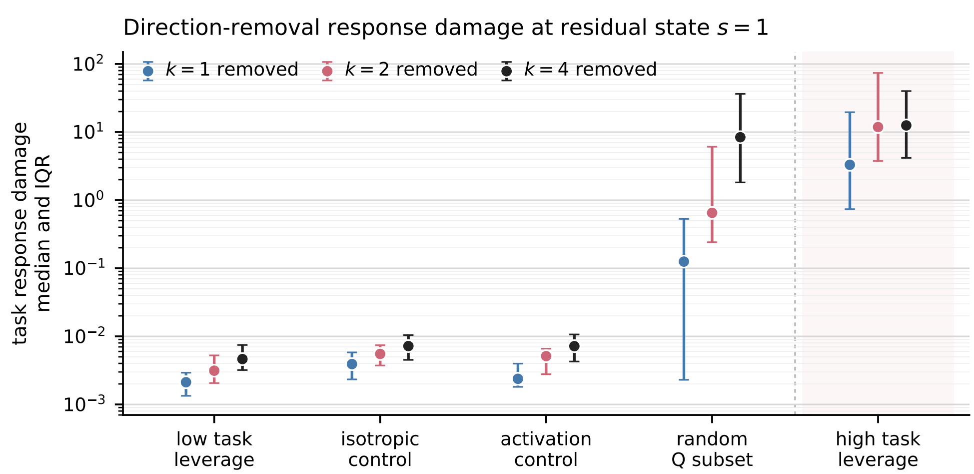

Figure 6: Experiment 3: selective causal effect of task-important activation directions. The plotted metric is task finite-difference response damage. High task-leverage removals cause orders of magnitude larger damage than matched low-task-leverage controls.

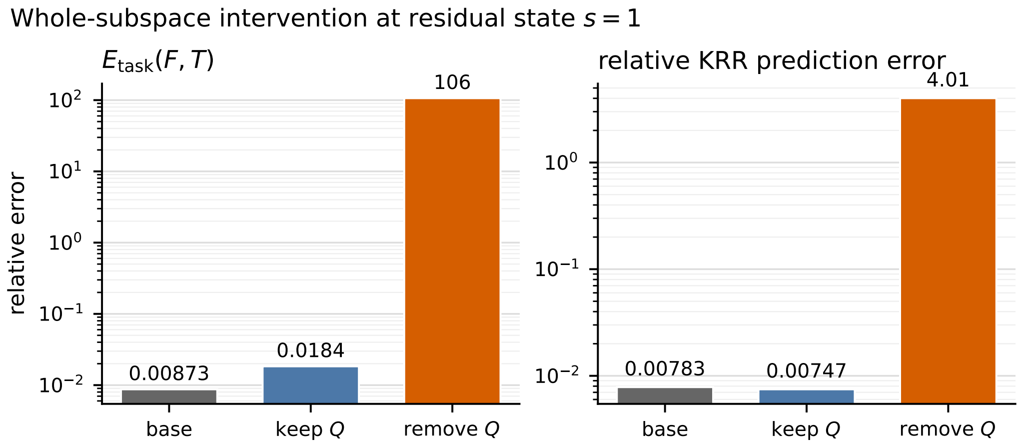

Figure 7: Experiment 3: whole-subspace complement ablation at residual state $s = 1$. Keeping only the Q-projection nearly preserves the task-local response; removing Q destroys it.

Together, the selective removals and complement ablation support the causal-use claim in an operational sense. The directions that the reduced KRR operator identifies as task-important are not merely decodable; preserving them largely preserves the task-local response, and removing them selectively destroys it.

9 What Is Justified

The experiments justify a bounded mechanistic claim:

> For the tested checkpoint, geometries, perturbation scale, and task-label probe distribution, the transformer's local label response is well approximated by a reduced KRR operator induced by a compact context-side activation subspace. In successful settings, layerwise response spans contain a prediction-risk-rank subspace whose induced reduced operator is prediction-risk sufficient; in the original linear setting, high-task-leverage directions from this subspace are causally important for the transformer's response.

They do not justify stronger claims. They do not show that the transformer globally equals ridge regression. They do not show that derivative matching at one sampled label vector implies equality on a neighborhood. They do not show that the transformer literally executes the explicit algorithm

$$y \mapsto QQ^\top y \mapsto K_t QQ^\top y$$

as its internal program. They also do not show that architecture alone guarantees the behavior. The prediction-risk rank calculation supplies a requirement for context-side subspaces; it is not an SGD theorem.

The clean final formulation is:

> The trained transformer locally realizes the ridge-regression label-response operator on task-distributed directions. In the tested regime, this behavior is explained by a compact context-side activation subspace whose induced reduced ridge operator is prediction-risk sufficient at approximately the prediction-risk rank. The layerwise diagnostic tracks when such a subspace becomes available; the causal intervention then asks whether the model actually uses the selected directions.

10 Logical Chain

The argument can be read as a sequence of commitments, each narrower than the previous one:

1. Freeze the episode geometry $G = (X_{\text{ctx}}, X_{\text{tgt}})$.
2. Under fixed G, ridge regression becomes $T = K_t A^{-1}$.
3. Treat the transformer as a label map $F : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}$.
4. Estimate the local response $D_\varepsilon F(y)[u]$.
5. Test behavior by checking $E_\varepsilon(F, T)$ on task-label probes.
6. Compute the prediction-risk rank r_T from $C_G = K_t A^{-1/2}$ and R_{KRR}.
7. Extract activation-derived context-side candidate spans C_ℓ.
8. Order each candidate span by KRR-targeted energy and form prefixes $Q_{\ell,k}$.
9. Construct the KRR-constrained reduced operator $T_Q = K_t Q Q^\top$.
10. Test subspace sufficiency with $\rho_G(T_Q)$.
11. Test model-subspace alignment with $E_\varepsilon(F, T_Q)$.
12. In causal runs, remove high-leverage Q-directions and check whether task-local response is damaged.

This is the full mechanism story: behavioral similarity, a KRR-structured activation certificate, prediction-risk-rank sufficiency, layerwise emergence of the relevant subspace, and causal specificity of high-leverage directions where the intervention has been validated.